SARDAR VALLABHBHAI PATEL INSTITUTE OF TECHNOLOGY, VASAD

Summer Internship Report

A PROJECT REPORT

Submitted by

Himanshu singh

220410101011

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In

Aeronautical Engineering

Sardar Vallabhbhai Patel Institute of Technology, Vasad

Gujarat Technological University, Ahmedabad

July 2025

NDC INSTITUTE OF AIRCRAFT MAINTENANCE ENGINEERING

NDC IAME
ESTABLISHED - 2008

Managed by : Naniben Damodardas Chokshi Educational & Medical Trust
Approved by DGCA, Ministry of Civil Aviation, Government of India,
Opp.Sumandeep Vidyapeeth, Waghodia Road, Vadodara - 391760,Gujarat,India.

Ref: NDCIAME/Cert/0725/1802

Date: 31/07/2025

TO WHOSOEVER IT MAY CONCERN

This is to certify that **Mr. HIMANSHU SINGH** studying B.E. in Aeronautical Engineering in 7th Semester bearing enrollment no. 220410101011 at Sardar Vallabhbahi Patel Institute of Technology, Vasad (Gujarat) has undergone 1 month of Industrial training at **NDC Institute of Aircraft Maintenance Engineering** located at Waghodia road, Vadodara (Gujarat) from 2nd July – 31st July' 2025.

During this period of training he has attended the practical training on *Aircraft Structure, Sheet Metal cutting and riveting techniques, Aircraft System, Propulsion System and Aircraft Avionic System.*

We have observed that the candidate is having hard working nature and sincere approach during the training period and wish him for better prospects in studies as well in career.

Chetan Nayak

(Chief Instructor/Training Manager)

NDC INSTITUTE OF AME
* VADODARA *

Sardar Vallabhbhai Patel Institute of Technology, Vasad

DECLARATION

I hereby declare that the Internship / Project report submitted along with the Internship / Project entitled **Industrial Internship in Aircraft Systems, Structures and, Avionics** submitted in partial fulfillment for the degree of Bachelor of Engineering in Aeronautical Engineering to Gujarat Technological University, Ahmedabad, is a Bonafide record of original project work carried out by me at NDC Institute of Aircraft Maintenance Engineering under the supervision of Pintu Maurya and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of Student	**Signature of the Student**
Kavya Nathabhai Gojiya	

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<u>Acknowledgement</u>

I would like to express my profound gratitude to the many individuals who made my one-month internship at the NDC Institute of Aircraft Maintenance Engineering a truly invaluable and enriching experience.

My sincere thanks go to the management and staff of the **NDC Institute of Aircraft Maintenance Engineering** for providing me with this incredible opportunity. I am especially thankful to my mentors and trainers, whose invaluable guidance, patience, and readiness to share their expertise in both documentation and practical work were instrumental to my learning. They provided the hands-on experience and knowledge that brought my theoretical studies to life.

I am also grateful to the **Gujarat Technological University (GTU)** and the faculty of my college for their continuous support and for structuring a curriculum that emphasizes the importance of practical training.

Finally, I wish to thank my family and friends for their unwavering support and encouragement throughout this journey.

Abstract

This report documents the experience and learning outcomes of a one-month summer internship undertaken at the **NDC Institute of Aircraft Maintenance Engineering** in Vadodara. The internship was a comprehensive program designed to provide a balanced exposure to both the theoretical and practical aspects of aircraft maintenance.

The first half of each day was dedicated to documentation and research, which involved creating **PowerPoint presentations on various aircraft systems and subsystems**. This work significantly enhanced my technical knowledge and improved my skills in organizing and communicating complex information. The second half of the day was focused on hands-on practical training. For the first fifteen days, I gained proficiency in modifying a workbench and learned the intricate skills of **stripping and tuning aircraft wires**. The subsequent fifteen days were dedicated to foundational mechanical work, including **drilling and riveting**.

A key highlight of the internship was gaining hands-on experience with a **Learjet aircraft**, along with observing a live **landing gear retraction and extension test**. I was also able to get a full understanding of the structural and system design of a **Cessna wing aircraft**. This real-world exposure successfully bridged the gap between academic theory and industry application, equipping me with a solid foundation for my future career in aircraft maintenance.

Weekly Overview of Internship Activities

Table (i)

1st WEEK	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	2/7/2025	Wednesday	Introduction of organization.
	3/7/2025	Thursday	Introduction to Electric Workbench and Modification.
	4/7/2025	Friday	Electric Workbench and Modification.

Table (ii)

2nd WEEK	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	7/7/2025	Monday	Electric Workbench and Modification.
	8/7/2025	Tuesday	Striping and Shouldering of wire.
	9/7/2025	Wednesday	Striping and Shouldering of wire.
	10/7/2025	Thursday	Electric Workbench and Modification and Striping and Shouldering of wire.

	11/7/2025	Friday	Electric Workbench and Modification and Striping and Shouldering of wire.

Table (iii)

3rd WEEK	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	14/7/2025	Monday	Introduction to Mechanical Shop
	15/7/2025	Tuesday	Sheet Metal Cutting and Drilling
	16/7/2025	Wednesday	Sheet Metal Cutting and Drilling
	17/7/2025	Thursday	Sheet Metal Cutting and Drilling
	18/7/2025	Friday	Sheet Metal Cutting and Drilling

4th WEEK	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	21/7/2025	Monday	Sheet Metal Drilling and Riveting
	22/7/2025	Tuesday	Sheet Metal Drilling and Riveting
	23/7/2025	Wednesday	Sheet Metal Drilling and Riveting
	24/7/2025	Thursday	Sheet Metal Drilling and Riveting
	25/7/2025	Friday	Aircraft Familiarization

5^{rd} WEEK	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	28/7/2025	Monday	Introduction to Control Surface and Landing Gear
	29/7/2025	Tuesday	Cockpit overview and visit to jet engine shop
	30/7/2025	Wednesday	Avionics and Propulsion System Study
	31/7/2025	Thursday	General Interaction with Faculties

Chapter 1: Introduction

1.1 History

The NDC Institute of Aircraft Maintenance Engineering (NDCIAME**)** was established in **2008** under the Naniben Damodardas Chokshi Educational & Medical Trust in Vadodara, Gujarat. It is approved by the Directorate General of Civil Aviation (DGCA), Ministry of Civil Aviation, Government of India, and is one of the premier institutions in western India dedicated exclusively to the field of aircraft maintenance engineering.

Since its inception, the institute has focused on delivering high-quality education and practical training in aircraft maintenance, aligned with the industry's technical standards. It aims to bridge the gap between academic knowledge and real-world aviation maintenance practices.

NDCIAME is equipped with modern workshops, well-furnished laboratories, and actual aircraft components, including engines, avionics systems, and airframes for hands-on training. The institute has played a pivotal role in preparing students for careers in civil aviation, MROs (Maintenance, Repair & Overhaul), and aerospace industries.

Over the years, NDC has built a strong reputation for its rigorous training modules, DGCA-compliant curriculum, and dedicated faculty, making it a reliable destination for aspiring aircraft maintenance professionals.

1.2 Mission & Vision

NDC Institute of AME mission is to make the Institute a 'Centre of Excellence' with highly developed educational infrastructure and excellent faculties as per international standards, provide aviation industry well trained & skilled manpower to create a pool of AME/Engineer's professionals' with world class competence by imparting exemplary education and training to add value to the society.

Become a leading institute to provide a transformative training to the students through rigorous course work and by providing an understanding of the needs of AME (Aircraft Maintenance Engineers) in Aviation Industry. An Institute that covers every aspect of theoretical & technical training requirements as per actual aircraft maintenance standards in aviation industry. Provide every individual an environment to achieve his/her career goals.

1.3 Hanger and Aircrafts

A newly constructed spacious hanger is available for accommodating available aircrafts. Various ground support equipment's like Electrical GPU, Hydraulic GPU, Air Compressor, Hydraulics jacks, Trestle and hoisting crane are available in the hanger to perform the maintenance training in efficient manner.

- Cessna 150 Fitted with Continental O-200 engine.
- Learjet 23 Fitted with General Electric CJ610 engines

1.4 Lab & Shops

The Institute has very well-equipped workshops and lab facilities and students are encouraged to acquire application skills and practically verify the knowledge acquired in the classroom. Labs and workshops are amongst the best equipped in the country. The basic training workshops and/or maintenance facilities separate from training classrooms are well designed for practical instruction appropriate to the planned training course. Workshop arrangements, laboratory layout and equipment installation are so designed as to totally facilitate the training process. Adequate electric supplies, compress air and other such facilities are provided as necessary.

Consistent with the requirements stated in CAR 147 (Basic), following workshops are available to impart Practical Training to the students:

a. Basic Metal Workshop

This is where you learn the foundational skills of working with various metals. You'd get hands-on experience with **measuring, marking, cutting, filing, and bending metal sheets and bars**. This shop teaches you how to use hand tools like hacksaws, files, shears, and bending machines. It's essential for understanding material properties and basic fabrication techniques, which are crucial for minor repairs and component manufacturing in aviation.

b. Machine Shop

In the machine shop, you'd move beyond hand tools to precision machining. Here, you'd learn to operate various machine tools like **drilling machines, and grinding machines**. This is where you'd practice creating precise components, shaping metal parts, and understanding tolerances. Skills learned here are vital for fabricating custom parts or repairing existing ones that require high accuracy.

c. Welding Shop

The welding shop is dedicated to teaching various welding techniques used in aircraft maintenance. You'd learn about different welding processes such as **TIG (Tungsten Inert Gas), MIG (Metal Inert Gas), and arc welding**, focusing on the specific metals and safety procedures relevant to aviation. This includes understanding weld integrity, identifying defects, and performing repairs on structural components or exhaust systems where welding is permitted.

d. Sheet Metal Shop

This shop is specialized for working with thin metal sheets, which form a large part of an aircraft's skin and internal structures. You'd gain practical experience in **cutting, forming, bending, and riveting sheet metal**. This includes learning about different types of sheet metal (like aluminum alloys and titanium), various fasteners, and techniques for repairing dents, cracks, and corrosion on aircraft surfaces. Your experience with drilling and riveting would have been a core part of this lab.

e. Engine Shop (Piston & Jet Engine)

This is one of the most critical shops for AME students.

- **Piston Engine:** You'd learn the internal workings of piston engines, including **disassembly, inspection, repair, and reassembly of components** like cylinders, pistons, crankshafts, and valve trains. You'd also study ignition, fuel, and lubrication systems specific to piston aircraft.
- **Jet Engine:** Here, you'd delve into the complexities of jet engines (turbofan, turbojet). Training involves **disassembly and assembly of engine modules**, inspection of turbine blades, compressor stages, combustion chambers, and understanding fuel, oil, and air systems. This shop is crucial for understanding engine performance, troubleshooting, and maintenance procedures. NDCIAME specifically mentions being approved for **Heavy Aeroplane, Jet Engine, Light Aeroplane & Piston Engine** training, indicating strong facilities in this area.

f. Electrical Shop

The electrical shop is where you'd get hands-on with aircraft electrical systems. You'd learn about **wiring, circuit testing, component installation, troubleshooting electrical faults, and working with aircraft batteries**. This includes understanding electrical schematics, using multimeters and other test equipment, and practicing wire stripping, crimping, and soldering techniques, which aligns with your internship experience.

g. Instrument Shop

This shop focuses on the maintenance and calibration of aircraft instruments. You'd learn about **flight instruments (e.g., altimeters, airspeed indicators), engine instruments, and navigation instruments**. This involves understanding their principles of operation, calibration procedures, troubleshooting malfunctions, and proper handling of delicate gauges and sensors.

h. Electronics and Radio Navigation Shop

This lab is dedicated to the complex electronic systems and communication/navigation equipment on an aircraft. You'd work with **avionics components, radio systems (VHF, HF), navigation systems (GPS, VOR, ILS), radar systems, and communication equipment**. This involves understanding signal processing, troubleshooting electronic circuits, and performing functional checks and repairs on these sophisticated systems.

i. Composite Material Lab

With the increasing use of composite materials in modern aircraft (like carbon fiber and fiberglass), this lab is becoming increasingly important. You'd learn about the **properties, fabrication, and repair techniques for composite structures**. This includes understanding different types of composites, methods for identifying damage, and techniques for patching and bonding composite materials.

j. NDT Lab (Non-Destructive Testing Lab)

The NDT lab is where you learn how to inspect aircraft components for defects without causing any damage to the part. You'd be trained in various NDT methods such as **dye penetrant inspection, magnetic particle inspection, eddy current testing, and ultrasonic testing**. These techniques are vital for detecting hidden cracks, corrosion, and other flaws in critical structural components and engine parts.

k. Battery Labs (Lead-Acid Battery & Ni-Cad Battery)

These specialized labs focus on the maintenance, charging, and testing of aircraft batteries.

- **Lead-Acid Battery:** You'd learn about the construction, charging procedures, capacity testing, and common maintenance issues like sulfation for lead-acid batteries, which are often used for general aviation and as backup power.
- **Ni-Cad Battery (Nickel-Cadmium Battery):** This lab would cover the specific characteristics, charging cycles (including deep cycling), thermal runaway prevention, and maintenance requirements unique to Ni-Cad batteries, which are widely used in larger commercial aircraft due to their high power output and reliability.

2. Internship Experience and Work Performed

This section uses the details you provided to create a concise summary of your daily work.

2.1 Duration and Schedule

My summer internship at the **NDC Institute of Aircraft Maintenance Institute** lasted for a period of one month, from [Start Date] to [End Date]. The daily schedule was structured into two distinct halves, focusing on both theoretical and practical aspects of aircraft maintenance. This dual-focus approach was highly effective, as it allowed for the immediate application of classroom knowledge in a hands-on environment.

2.2 Work Performed - First Half of the Day

The first half of each day was dedicated to **documentation and theoretical learning**. My primary task was to create detailed **PowerPoint presentations on various aircraft systems and subsystems**. This involved extensive research into topics such as **pneumatic systems, hydraulic systems, and flight controls**. The process of developing these presentations required me to synthesize complex technical information from maintenance manuals and industry publications into a clear and educational format. This work not only deepened my understanding of these systems but also honed my skills in technical communication and data organization, which are essential for an Aircraft Maintenance Engineer.

2.2.1 Air Conditioning and Cabin Pressurization System

The presentation I created during my internship on "Air Conditioning and Cabin Pressurization" provides a comprehensive and detailed look at the Environmental Control System (ECS) of an aircraft. It's a multi-faceted system that is absolutely essential for the safety and comfort of everyone on board.

The presentation likely broke down the topic into several key sections, starting with the fundamental purpose and moving into the intricate mechanics of how it all works:

1. The Importance of Air Conditioning and Pressurization

The presentation begins by establishing the critical need for these systems. It explains that at cruising altitudes (e.g., 35,000 feet), the outside air is too cold and has too little oxygen to sustain life. The presentation would cover:

- The physiological effects of low pressure and temperature.
- The primary objective: maintaining a stable, safe, and comfortable "cabin altitude" (typically around 8,000 feet) and a pleasant temperature regardless of outside conditions.

2. Air Supply: Where the Air Comes From

This section details the various sources of pressurized air, which is the foundation of the entire system.

- **Engine Bleed Air (Primary Source):** The presentation explains that during flight, air is "bled" off from the compressor section of the aircraft's jet engines. This air is hot, at high pressure, and is the primary source for the air conditioning and pressurization systems.
- **Auxiliary Power Unit (APU):** It describes how the APU, a small jet engine in the tail of the aircraft, serves as an independent source of bleed air when the main engines are off or for specific operations on the ground.
- **Ground Air Carts:** The presentation also covers the use of external carts on the ground, which provide conditioned air to the cabin, saving fuel and reducing wear on the APU and main engines.

3. The Air Conditioning Process: The Air Cycle Machine (ACM)

This is the core technical part of the presentation, focusing on how hot bleed air is transformed into cold air. It would have explained the system in detail, following the principles of the **Reverse Brayton Cycle**:

- **Compression:** The hot bleed air is first compressed further by a compressor within the Air Cycle Machine (ACM), which raises its temperature and pressure.
- **Primary Cooling:** This super-hot air then passes through a primary **heat exchanger** (essentially a radiator), where it is cooled by a ram air fan using outside air.
- **Expansion and Cooling:** The cooled air is then forced through a **turbine**. As the air expands and does work on the turbine, its temperature drops dramatically—often below freezing.
- **Water Separation:** Because of this rapid cooling, any moisture in the air condenses. A **water separator** removes this moisture, preventing ice from forming in the ducts.
- **Temperature Regulation:** Finally, the super-cold air is mixed with a controlled amount of hot bleed air via a bypass valve to achieve the precise temperature requested by the crew before it enters the cabin.

4. Cabin Pressurization: Keeping the Cabin Sealed

This part of the presentation focuses on how the cabin pressure is regulated. It explains that the aircraft is essentially a pressurized vessel.

- **Inflow and Outflow:** The system works by controlling the amount of air that flows out of the cabin, relative to the constant inflow from the ECS packs.
- **The Outflow Valve:** The presentation highlights the **outflow valve** as the most critical component. By adjusting how wide this valve is open, the system maintains a specific pressure differential between the inside and outside of the aircraft.

- **Pressurization Controller:** It describes the automated pressurization controller that monitors altitude and automatically adjusts the outflow valve to maintain the desired cabin altitude.

In summary, the presentation would serve as a complete guide, from the basic physics of air pressure to the intricate engineering of the system's components, providing a solid theoretical foundation for any AME student.

2.2.2 Pneumatic Systems

The presentation provided a detailed technical overview of aircraft landing gear, a mission-critical system essential for safe ground operations, from taxiing to takeoff and landing. The content is structured to cover the fundamental purpose, key components, operational mechanics, and critical safety features.

1. Core Functions and Design Philosophy

The presentation begins by establishing the primary functions of the landing gear beyond simply supporting the aircraft on the ground. It highlights its roles in:

- **Weight Bearing:** Supporting the entire weight of the aircraft, including its payload, during ground operations.
- **Shock Absorption:** Dissipating the immense energy of landing impact to prevent structural damage to the airframe.
- **Directional Control:** Allowing for steering and braking during taxi and takeoff rolls.
- **Drag Reduction:** Explaining the aerodynamic importance of retractable landing gear, which tucks away during flight to reduce drag and improve fuel efficiency.

2. Main Components and Structures

A significant portion of the presentation focuses on the individual components that make up the landing gear system. These include:

- **Oleo Strut (Air-Oil Shock Absorber):** This is the heart of the system. The presentation would have explained how a cylinder filled with nitrogen (or another inert gas) and hydraulic fluid works to compress and absorb landing shocks.
- **Side and Drag Braces:** These structural members are crucial for keeping the gear aligned and supporting it against forces during takeoff and landing.
- **Uplocks and Downlocks:** The presentation details these mechanical locking mechanisms. Uplocks secure the gear in the retracted position, while downlocks ensure the gear is firmly in place for landing, preventing it from collapsing under the aircraft's weight.
- **Wheel and Brake Assembly:** It covers the high-performance wheels and multi-disc carbon brakes designed to handle the high speeds and weight of the aircraft. This section also touches upon anti-skid systems that prevent tire skidding during braking, similar to those in a car.
- **Steering System:** For the nose gear, it explains the different methods of steering, such as hydraulic power steering controlled by a tiller in the cockpit, which allows for precise ground maneuvering.

3. The Operational Cycle

The presentation provides a step-by-step breakdown of the hydraulic retraction and extension sequence, including:

- **Retraction:** The pilot selects the "gear up" position, which commands the hydraulic system to release the downlocks, retract the gear into the fuselage or wings, and finally engage the uplocks to secure it.
- **Extension:** The process is reversed, with hydraulic pressure releasing the uplocks, lowering the gear, and then applying pressure to ensure the downlocks are engaged. The presentation emphasizes the importance of a cockpit indication, often three green lights, to confirm that all gear is "down and locked."

4. Safety and Emergency Systems

This section is vital, as it covers the redundancies built into the system to ensure safety. The presentation outlines:

- **Emergency Extension:** Explaining the backup systems, such as a **gravity drop** or a **hand crank**, that a pilot can use to lower the landing gear if the primary hydraulic system fails.
- **Proximity Switches and Squat Switch:** It explains how sensors determine the position of the gear (up, down, in transit) and how the "squat switch" on the main landing gear prevents accidental retraction while the aircraft is on the ground.
- **Warning Systems:** The presentation would have covered the various aural and visual warnings that alert the crew if the gear is not in the correct configuration for landing.

This comprehensive overview demonstrates a deep understanding of not only the components but also the operational and safety-critical context in which they function.

2.2.3 Landing Gears

The presentation provided an in-depth look at the aircraft pneumatic system, a fundamental utility system that supplies pressurized air to power a variety of other critical aircraft systems. It demonstrated how a single source of high-pressure air is carefully managed and distributed to ensure safe and efficient flight operations.

1. Purpose and Function

The presentation would begin by defining the core role of the pneumatic system: to act as a source of pressurized air. This air, often referred to as **bleed air**, is a valuable resource used for a range of functions across the aircraft. The system's primary goal is to provide this air at the correct pressure and temperature for each specific application, ensuring reliability and safety.

2. Sources of Pressurized Air

This section details where the pneumatic system gets its air. The presentation would cover the various sources, highlighting their roles in different flight phases:

- **Engine Bleed Air (Primary Source):** This is the main source of pressurized air during flight. Air is taken, or "bled," from the compressor section of the aircraft's running jet engines. This air is extremely hot and at high pressure.
- **Auxiliary Power Unit (APU):** The presentation explains how the APU, a small turbine engine in the aircraft's tail, is a secondary source of bleed air. It is primarily used on the ground to provide air for air conditioning and engine starting, but can also be used in-flight as a backup.
- **Ground Air Carts:** For ground operations, especially in maintenance, external pneumatic carts can be connected to the aircraft to supply a temporary source of pressurized air, preventing the need to run the APU or main engines.

3. Key Components and Distribution

This part of the presentation breaks down the physical architecture of the pneumatic system:

- **Bleed Air Ducts and Manifolds:** A network of specialized, high-temperature ducts and manifolds routes the bleed air from its sources to its various destinations.
- **Pressure Regulating Valves (PRVs):** Since bleed air is supplied at very high pressure, PRVs are used to reduce it to a safe and usable pressure for downstream systems.
- **Shutoff Valves:** These valves are critical for safety, allowing the crew or a control system to isolate sections of the pneumatic system to manage leaks or prevent over-pressurization.
- **Cross-Bleed Valve:** A crucial component that allows air from one engine's pneumatic system to be routed to the other side of the aircraft. This is essential for engine starting and provides redundancy in case of a bleed air source failure.
- **Overheat and Leak Detection:** The presentation would have described the sensors placed along the ducting that monitor for leaks or excessively high temperatures, providing warnings to the flight crew.

4. Major Applications of Pneumatic Air

The presentation would conclude by detailing the diverse systems that depend on the pneumatic system for their operation. This section illustrates the central importance of bleed air to the aircraft's functionality:

- **Environmental Control System (ECS):** The most significant consumer of bleed air. This air is used by the Air Cycle Machines to provide air conditioning and pressurization for the cabin and cockpit.
- **Engine Starting:** For certain aircraft types, bleed air from the APU or a running engine is directed to a starter turbine to spin up a non-operating engine during the start sequence.
- **Wing and Engine Anti-Icing:** Hot bleed air is channeled through small holes on the leading edges of the wings and engine cowls to heat the surfaces and prevent ice formation.
- **Hydraulic System Reservoirs:** Pneumatic pressure is used to pressurize the hydraulic fluid reservoirs, ensuring a constant and stable supply of fluid to the hydraulic pumps.
- **Water and Waste Systems:** Pressurized air can also be used to move water from the supply tanks and to assist in the flushing of the aircraft's lavatory systems.

This detailed overview provides a solid foundation for understanding the pneumatic system's role as the central air utility provider, demonstrating its critical connection to other essential aircraft systems.

2.3 Work Performed - Second Half of the Day

The second half of the day was focused on **hands-on practical work**, providing a direct application of the theoretical knowledge gained. This period was further divided into two distinct fortnightly segments to cover a broad range of skills:

- **First 15 Days: Electrical and Benchwork** During this period, I was involved in **modifying a work bench** to improve its efficiency and organization. This task taught me about optimizing a workspace for specific maintenance procedures. A key learning experience was gaining proficiency in **stripping and tuning of aircraft wires**, a fundamental skill in aircraft electrical maintenance. I learned about the specialized tools and techniques required to handle delicate aircraft wiring, including the use of wire strippers, crimping tools, and proper wire termination. This included understanding the importance of wire gauge, insulation, and the prevention of damage during handling.
- **Next 15 Days: Mechanical Work and Aircraft Observation** The final two weeks were dedicated to **mechanical work** and gaining practical insight into aircraft systems. The core activities included **drilling and riveting**, which are crucial processes in aircraft structural repair and assembly. I learned proper drilling techniques to ensure hole integrity, how to select the correct types of rivets for different applications (e.g., solid rivets, blind rivets), and the importance of precision in structural work to maintain airworthiness standards.

  In addition to these hands-on skills, I was fortunate to gain practical experience on a **Learjet aircraft**, working on its systems. I also had the opportunity to **witness a landing gear retraction and extension test**, which provided a firsthand look at the hydraulic and mechanical systems in operation. Furthermore, I was able to get a comprehensive understanding of the structure and systems of a **Cessna wing aircraft**, observing its design and maintenance principles. This experience provided a tangible understanding of the quality control and meticulous attention to detail required in mechanical maintenance and system checks.

2.3.1 Electrical Workbench Modification

What is an Electrical Workbench?

An electrical workbench is the central hub for anyone working with electronics, from hobbyists to professional engineers. It is a dedicated, organized, and properly equipped workspace where components are tested, circuits are built and repaired, and projects are brought to life. More than just a table, it is an ecosystem of tools and equipment designed to facilitate precise work while ensuring both the safety of the user and the integrity of sensitive components.

Essential Components of an Electrical Workbench

A complete workbench consists of several key categories of tools and equipment, each serving a specific purpose.

1. The Workbench and Anti-Static Protection

- **Sturdy Surface:** The foundation is a solid, stable table or bench. The surface material is important; a **non-conductive** and easy-to-clean surface is ideal.
- **Grounding Mat:** An **Electrostatic Discharge (ESD)** mat is a soft, conductive mat placed on the work surface. It is connected to a ground point to safely dissipate static electricity, preventing damage to delicate components like microcontrollers and integrated circuits.
- **ESD Wrist Strap:** This strap is worn by the user and connects to the grounding mat or a dedicated ground point. It constantly equalizes the user's electrical potential with the workbench, ensuring that no static charge can be transferred to sensitive components.

2. Power and Prototyping

- **Variable DC Power Supply:** A must-have for testing. This device provides a stable, adjustable DC voltage and current, allowing you to power circuits with the precise requirements they need, from a few volts for a small sensor to higher voltages for motors.
- **Breadboards and Prototyping Boards:** These are essential for quickly and non-permanently assembling circuits. **Breadboards** are used for temporary, solderless connections, while **prototyping boards** allow for more permanent circuits using through-hole components.

3. Measurement and Analysis Tools

- **Digital Multimeter (DMM):** This is the most fundamental tool on the bench. It measures voltage, current, and resistance and is indispensable for troubleshooting circuits and verifying connections.
- **Oscilloscope:** A more advanced tool for visualizing electrical signals over time. It is used to analyze waveforms, check signal integrity, and troubleshoot complex digital and analog circuits.
- **Function Generator:** This tool produces various electronic waveforms (sine, square, triangle, etc.) at different frequencies and amplitudes. It is used to inject signals into a circuit to test its response.

4. Hand Tools and Assembly

- **Soldering Iron or Station:** An essential tool for creating permanent electrical connections. A soldering station with adjustable temperature control is highly recommended for working with different types of solder and components.
- **Wire Strippers and Cutters:** Specialized tools for safely preparing wires. Wire strippers remove insulation without damaging the conductor, and flush cutters are used to trim component leads cleanly.
- **Pliers and Tweezers:** A variety of small pliers (needle-nose, flat-nose) and tweezers are used for holding, manipulating, and placing tiny components.

- **Screwdriver Set:** A complete set of small screwdrivers, often with interchangeable bits, is necessary for assembling enclosures and working with terminal blocks.

5. <u>Organization and Safety</u>

- **Good Lighting and Magnification:** A well-lit workspace with a bright, adjustable lamp is critical. For fine detail work, a magnifying lamp or a stereomicroscope can prevent eye strain and improve precision.
- **Component Storage:** Organizers with small drawers or compartments are vital for storing and categorizing resistors, capacitors, and other small parts, keeping the workspace tidy and efficient.
- **Fume Extractor:** When soldering, a **fume extractor** with a carbon filter is used to draw away and filter the harmful fumes produced by the flux in the solder.
- **Safety Glasses:** Eye protection should always be worn when soldering, cutting wires, or performing any task where small debris could be a hazard.

➢ <u>Best Practices for a Safe and Effective Workbench</u>

Beyond the tools themselves, a few key practices define a great electrical workbench:

- **Organize Your Tools:** Keep your most-used tools within easy reach and everything else neatly stored. This saves time and prevents clutter.
- **Keep It Clean:** Regularly clear your workspace of old components, wire clippings, and debris. This helps prevent short circuits and makes it easier to find what you need.
- **Always Be Grounded:** Before touching any static-sensitive components, always put on your ESD wrist strap and ensure it is properly connected.
- **<u>Read the Manuals:</u>** Understand the tools you are using, especially high-voltage power supplies and oscilloscopes, to avoid damaging the equipment or yourself.

2.3.1.A Components of Electrical Workbench

An electrical workbench is designed to test a wide range of things, including:

- **Individual components:** You can measure the properties of individual parts like a resistor's value, a capacitor's capacitance, or a diode's forward voltage drop using a **multimeter**.
- **Circuit functionality:** You can verify that a complete circuit path exists using the multimeter's **continuity test**.
- **Power and voltage:** A **variable DC power supply** is used to provide the correct voltage and current to a circuit, allowing you to power it up and test its behavior. You can also measure the actual voltage at different points in the circuit with a multimeter.
- **Electrical signals and waveforms:** An **oscilloscope** is a powerful tool for visualizing signals. It lets you analyze the shape, frequency, and amplitude of a signal to see if it's behaving as expected or if there is noise or distortion.
- **Circuit response:** A **function generator** is used to inject specific types of signals (like sine waves or square waves) into a circuit. You can then use the oscilloscope to see how the circuit modifies or responds to that signal, which is crucial for testing filters, amplifiers, and oscillators.

Essentially, the workbench's tools allow you to power, probe, and analyze a circuit from every angle to ensure it is functioning correctly.

- **Equipment list**

1. Motor
2. Voltage (PHASE A,B AND C)
3. Alternator display
4. Load (1,2,3,4,5 and 6)
5. Neutral
6. Rectifier unit
7. Start

8. Dc gen display
9. Monitor (Starter)
10. Alternator exciter
11. Transformer secondary
12. Induction motor testing
13. Power monitoring display
14. Transformer testing
15. CT
16. AC (0-220V)
17. Main switch (ac)
18. Main switch (dc)
19. Multimeter
20. Galvanometer
21. Inductor testing
22. Ammeter(dc)
23. EMF
24. EMF gen testing
25. Electromagnet testing
26. Voltmeter (dc)

2.3.1.B Modifications Made

- **Modifications made**

1. LCD
2. CT
3. Power monitoring display
4. Monitor (stator)
5. Variac starts sparking above 200V
6. Alternator display
7. Key orientation
8. Fixing the front board with wooden board

9. Putting up name tags
10. Mentioning warnings

2.1.2 Wire Stripping, Soldiering and Crimping

At NDC Institute of Aircraft Maintenance Engineering, we were trained in wire stripping and soldering techniques commonly used in aircraft electrical systems. We learned to safely remove insulation from wires without damaging the conductor. The stripped wires were then soldered to terminals and connectors to ensure reliable electrical joints. This activity improved our hand skills and introduced us to industry-standard practices for wiring in aviation. Also, were taught how to apply wire terminators and safety in the process of crimping.

2.3.2 Basics of Mechanics

This section focuses on fundamental mechanical skills used for joining metal parts. Drilling creates holes, and riveting uses those holes to permanently fasten pieces together.

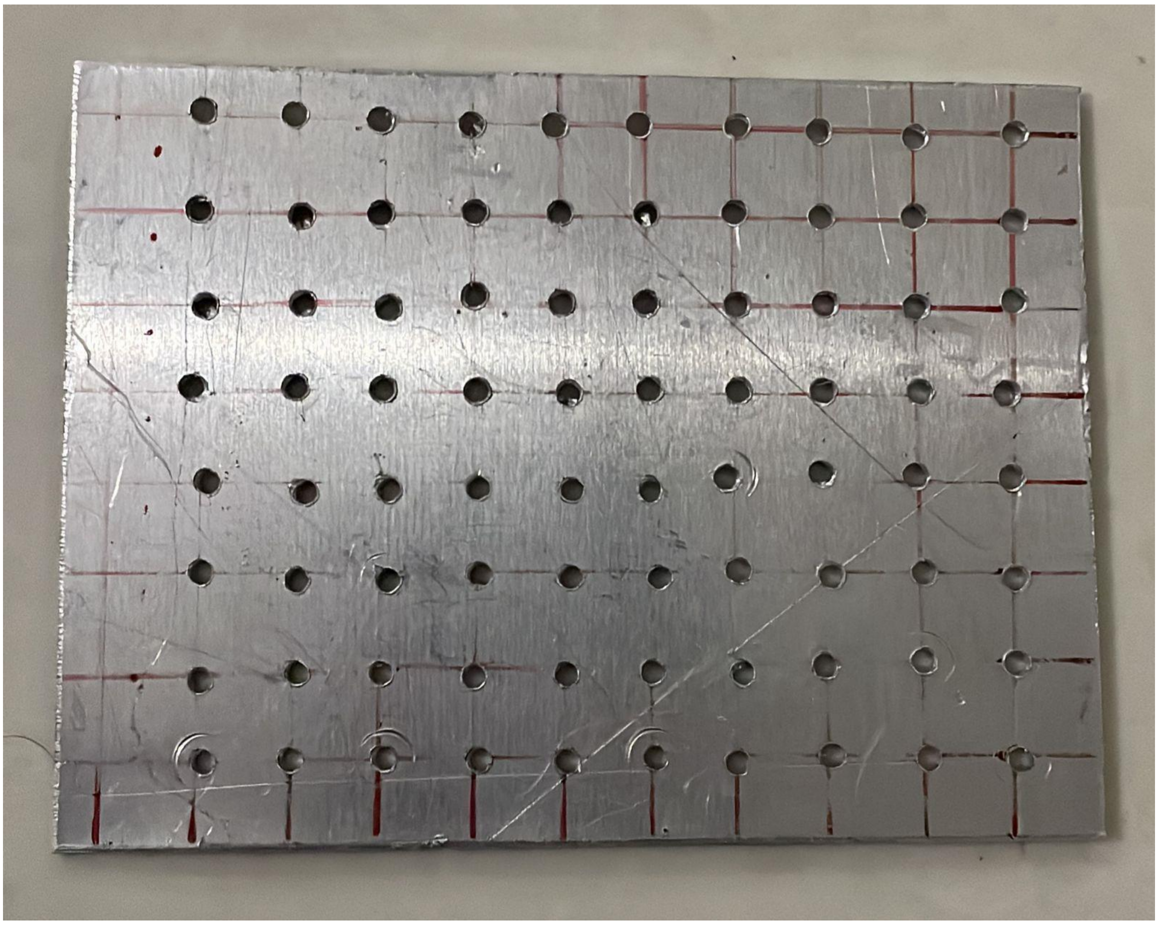
2.3.2.A Sheet Metal Cutting

During our internship at NDC Institute of Aircraft Maintenance Engineering, we learned the fundamentals of the sheet metal cutting process, which is essential in aircraft structural fabrication. We practiced cutting aluminum sheets to specified shapes and sizes using both hand and powered tools. The process required accurate measurements, clean cuts, and proper safety practices to ensure the quality and precision of components

2.3.2.B Drilling Using a 2.5mm Drill Bit

- **Explanation:** This task involves using a drill press or a hand drill to create a small, precise hole in a metal sheet. The 2.5mm drill bit is selected for its small diameter, which is useful for creating pilot holes or for joining components with small rivets. Before drilling, the exact location of the hole is marked with a center punch to prevent the drill bit from "walking" or slipping on the metal surface.

- **Practical Experience:** The process begins with clamping the metal sheet firmly to the workbench. The center punch makes a small indentation, which feels like a crisp, definite tap. As the drill press is turned on, a whirring sound fills the room. When the drill bit makes contact with the metal, a satisfying scraping sound starts. Applying steady, even pressure is key. Too much pressure can break the small, delicate bit, while too little pressure will just dull it. As the drill cuts through the metal, you can feel a subtle change in resistance. Once the bit breaks through the other side, there is a sudden drop in pressure. The drill must be withdrawn carefully to avoid damaging the new hole.



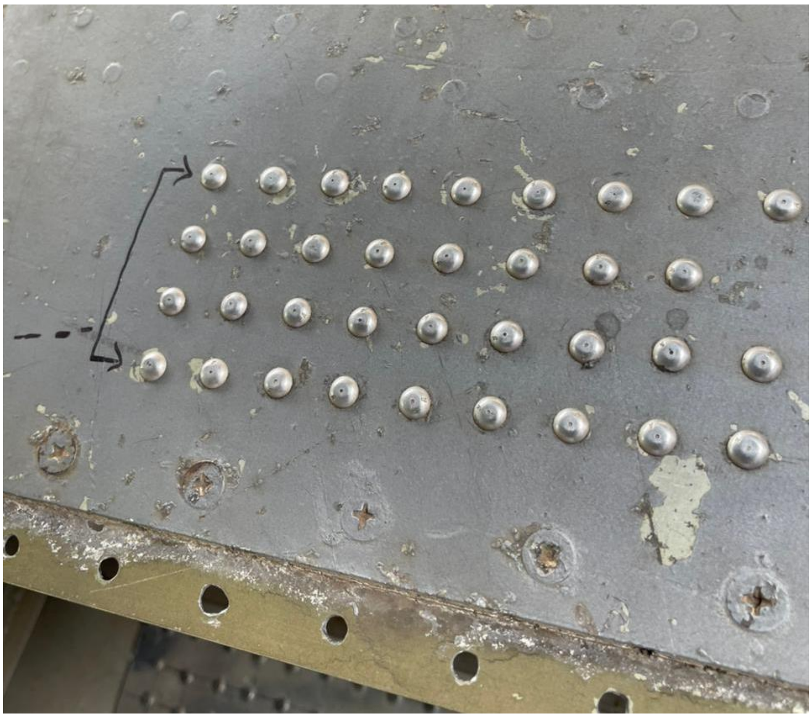
2.3.2.C Drilling Using a 3mm Drill Bit

- **Explanation:** Similar to the previous step, this involves drilling a hole, but with a slightly larger 3mm drill bit. This bit is used to widen an existing 2.5mm pilot hole or to create a hole for a standard 3mm rivet. Using a pilot hole first reduces stress on the larger drill bit and ensures the final hole is accurately placed.

- **Practical Experience:** The procedure is very similar to using the 2.5mm bit, but the experience feels a little different. The larger 3mm bit is more robust and less likely to snap, allowing for slightly more pressure. The sound of the drilling is a bit deeper, and the shavings that curl away from the hole are noticeably thicker. The resistance from the metal feels stronger and more consistent. If a pilot hole was used, the drill glides into the metal with less of the initial scraping and more of a steady, burrowing motion. The final result is a clean, slightly wider hole, ready for the next step.

2.3.2.D Riveting

- **Explanation:** Riveting is a permanent joining method that uses a rivet—a small metal pin—to fasten two or more sheets of metal together. A rivet gun pulls the shaft of the rivet, causing the body to deform and create a tight, secure bond.
- **Practical Experience:** This is the most physically satisfying part of the process. You align the two metal sheets and push the rivet through the hole. The fit should be snug but not so tight that you have to force it. Next, the rivet gun is placed over the rivet's head. Squeezing the handle of the rivet gun multiple times is a rhythmic, deliberate action. With each squeeze, you feel a distinct "clunk" as the gun pulls the rivet's shaft. The rivet gun makes a few of these clunking sounds, followed by a final, sharp "pop" as the shaft breaks off. When you remove the gun, the two pieces of metal are now a single, solid unit. The rivet head is deformed and flush with the surface, and the bond is strong enough that it cannot be undone without drilling out the rivet.



2.3.3 Aircraft Overview

2.3.3.A Aircraft Familiarization

During our internship at NDC, we were introduced to the Cessna 150 (a two-seat, single-engine trainer aircraft) and a Learjet (a twin-engine business jet) housed in the institute's hangar. The instructors provided a comprehensive briefing on various aircraft systems and their functions, helping us relate theoretical knowledge with real-world aircraft components.

1. Control Surfaces

We learned about the primary control surfaces: ailerons, elevators, and rudders. These surfaces are responsible for roll, pitch, and yaw respectively. We also observed secondary controls like flaps and trim tabs which help in stability and maneuvering, especially during take-off and landing.

2. Propulsion System

In the Cessna 150, we studied the reciprocating engine, which uses an air-cooled, horizontally opposed piston engine. For the Learjet, a turbofan engine was used, and we were shown the main components including the fan, compressor, combustion chamber, turbine, and nozzle. The basic working cycle and thrust generation principles were explained.

3. Cockpit Overview

The cockpit tour included familiarization with basic flight instruments such as the altimeter, airspeed indicator, attitude indicator, and engine gauges. The difference between analog gauges in the Cessna and more advanced avionics in the Learjet was highlighted.

4. Landing Gear Mechanism

We observed the tricycle-type fixed landing gear of the Cessna and the retractable landing gear of the Learjet. The systems included oleo struts, hydraulic actuators, wheels, and braking systems. We were briefed on how landing gear is extended and retracted and its role during taxiing, takeoff, and landing.

3. Learning Outcomes

This section translates your activities into a list of skills and knowledge you gained.

- **Comprehensive System Knowledge:** Developed a strong theoretical understanding of several aircraft systems and subsystems (e.g., pneumatic, hydraulic, flight controls) through the creation of technical presentations. This provided a foundational knowledge base for future studies and career.
- **Hands-On Electrical Proficiency:** Gained practical experience in aircraft electrical work, specifically with wire stripping and tuning. This included learning to use industry-standard tools and understanding the principles of proper wire handling.
- **Fundamental Mechanical Skills:** Acquired hands-on skills in basic aircraft structural work, including precision drilling and riveting. This skill set is vital for any AME and provides a strong foundation in structural maintenance.
- **Real-World System Observation:** Gained invaluable insight into the operation of a **Learjet** and a **Cessna wing aircraft**, including witnessing a **landing gear test**, which provided practical context to theoretical knowledge.
- **Workplace Efficiency:** Learned how to optimize a workspace through the modification of the work bench, which enhanced my understanding of an organized and efficient maintenance environment.
- **Research & Communication:** Improved my research skills and learned how to effectively present complex technical information in a clear and concise manner, a critical soft skill for an AME.

4. Conclusion

4.1 Summary of Experience

My one-month internship at NDC Institute of Aircraft Maintenance was an invaluable experience. The balanced curriculum of documentation and hands-on practical work provided a holistic view of the AME profession. From in-depth theoretical research to the practical skills of drilling, riveting, and wire maintenance, as well as hands-on experience with specific aircraft like the **Learjet** and **Cessna**, I was able to bridge the gap between classroom knowledge and real-world application.

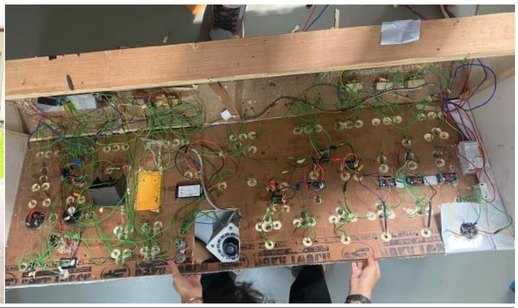
4.2 Future Recommendations

This internship has solidified my interest in aircraft maintenance and provided a clear direction for my future career. The skills and knowledge I gained will undoubtedly aid me in my final year of study and beyond. I am now more confident in my ability to perform practical tasks and understand the intricate systems that keep an aircraft airworthy.

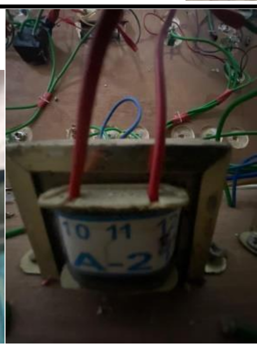
5. GALLERY









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